

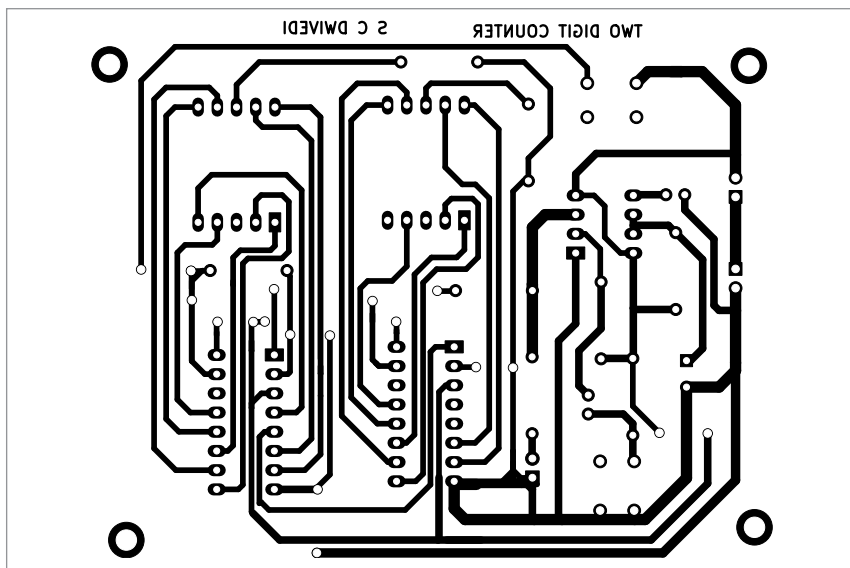


Fig. 3: Actual-size PCB layout

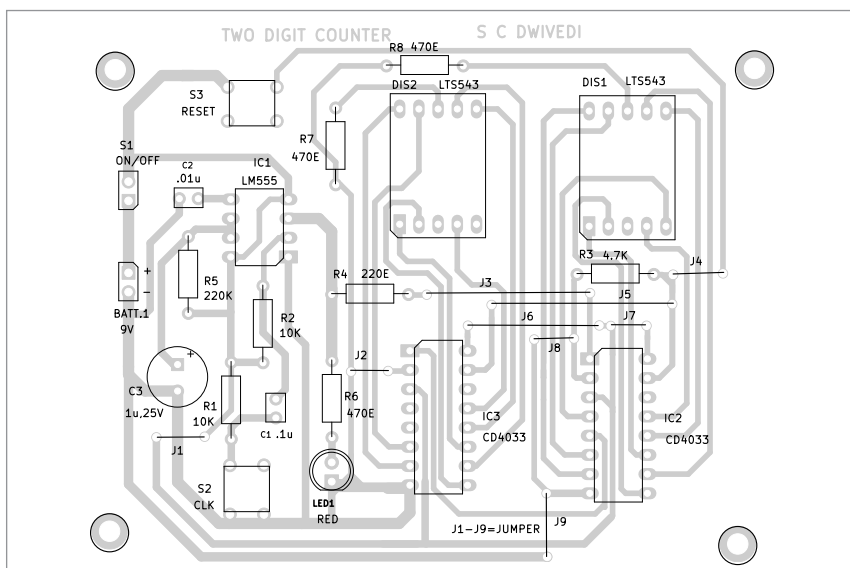


Fig. 4: Components side

7-segment display. The LM555 timer prevents signal debouncing from the clock generated by switch S1 or a suitable sensor, ensuring a clean and precise clock pulse.

The CD4033 IC is a decade counter that counts from 0 to 9. It features an inbuilt BCD output, providing the binary representation of numbers 0 to 9 on its output pins. The two CD4033 ICs are connected in a cascading configuration to

enable counting from 00 to 99.

The first CD4033 IC (IC2) manages the one's place. The clock pulse from the LM555 timer is fed into the clock input of IC2, causing it to increment by one for every pulse. The second CD4033 IC (IC3) controls the tens place. The carry-out pin of IC2 triggers IC3. When IC2 reaches 9 and resets to 0, it generates a carry-out pulse that triggers IC3 to increment the tens

digit by one.

Each CD4033 IC provides BCD outputs corresponding to its count value (0-9). These outputs are fed to the 7-segment displays. The output of IC2 connects to the first 7-segment display (DIS1), which shows the current count for the one's place. The output of IC3 connects to the second 7-segment display (DIS2), which shows the current count for the tens place.

When the device is powered on, pressing switch S2 momentarily triggers the LM555 timer, providing a clock pulse to IC2. Each pulse increments the count in the CD4033 ICs. IC2 increments the one's digit, and when it overflows (reaches 9), it generates a carry-out signal that increments the tens digit in IC3. As a result, the two 7-segment displays count sequentially from 00 to 99.

This device effectively functions as a simple digital counter, suitable for applications such as timing, object counting, and sequential numerical displays.

## Construction and testing

An actual-size, single-sided PCB layout for the two-digit counter is shown in Fig. 3, and its

### PARTS LIST

#### Semiconductors:

- IC1 - LM555 timer
- IC2, IC3 - CD4033 counter with 7-segment display
- LED1 - 5mm LED

#### Resistors (all 1/4-watt, $\pm 5\%$ carbon):

- R1, R2 - 10k $\Omega$
- R3 - 4.7k $\Omega$
- R4 - 220 $\Omega$
- R5 - 220k $\Omega$
- R6-R8 - 470 $\Omega$

#### Capacitors:

- C1 - 0.1 $\mu$ F ceramic disc
- C2 - 0.01 $\mu$ F ceramic disc
- C3 - 1 $\mu$ F, 25V electrolytic

#### Miscellaneous:

- BATT.1 - 9V battery
- S1 - SPST switch
- S2, S3 - Push-to-on switch
- DIS1, DIS2 - LTS543 common cathode 7-segment display